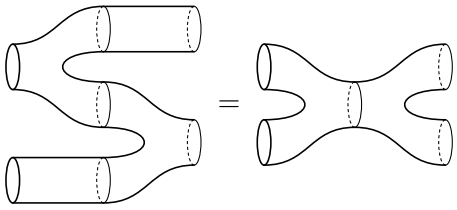
The plumber package

Diagrams of 2d TQFT cobordisms in LaTeX with Ti*k*Z

Version 1.1, 2026/10/27

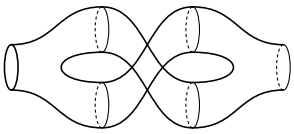
A two-dimensional topological quantum field theory is a symmetric monoidal functor out of the cobordism category 2Cob, whose morphisms are surfaces between disjoint unions of circles. Every such surface is a composite of a few generators (cylinders, pairs of pants, caps, …), and this package draws the composite from a description layer by layer:



Each `\PL{⟨m,n⟩}{⟨generators⟩}` is one layer, the tensor product of the listed generators, going from m to n circles; consecutive `\PL` commands are composed into one surface. The package keeps track of how many circles there are on every seam, places them so that the tubes run smoothly, and draws the seams where layers are glued as circles with a solid front and a dashed back. With `reduce` it computes the topology of the result and draws the normal form instead.

1 Quick start

```
\usepackage{plumber}   % loads TikZ
...
\PL{1,2}{pant}\PL{2,2}{twist}\PL{2,1}{copant}
```



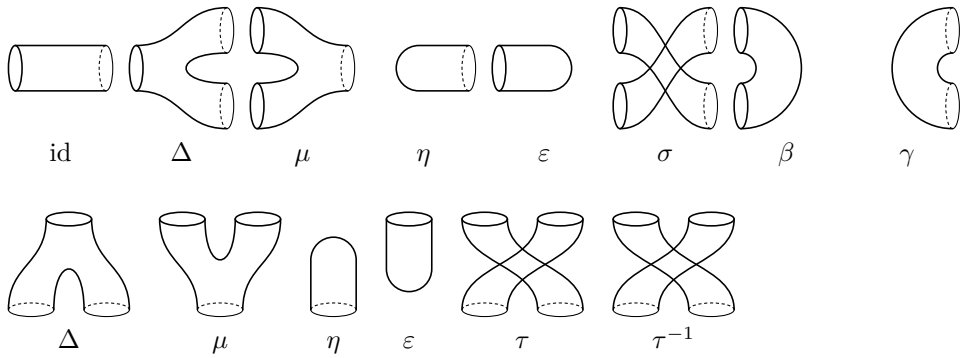
The first argument is the number of input and output circles of the layer (`2,3` means two circles in, three out); it is checked against the generators and may also be left empty. The second argument is a comma-separated list of generators, read from the first circle to the last (top to bottom in horizontal diagrams, left to right in vertical ones). `id^3` is short for `id, id, id`, and the same works for any generator.

2 Generators

name	aliases	circles	χ	symbol
`id`	`cyl`, `cylinder`	$1 \to 1$	0	id
`pant`	`pants`	$1 \to 2$	-1	Δ
`copant`	`copants`	$2 \to 1$	-1	μ
`unit`	`birth`	$0 \to 1$	1	η
`counit`	`death`	$1 \to 0$	1	ε
`swap`		$2 \to 2$	0	σ
`twist`		$2 \to 2$	0	τ
`itwist`		$2 \to 2$	0	τ^{-1}
`pair`	`ev`	$2 \to 0$	0	β
`copair`	`coev`	$0 \to 2$	0	γ

Diagrams are read in the direction of composition: from left to right, or from top to bottom with `vertical`. So `pant` splits one circle into two and `copant` merges two circles into one. `swap` exchanges two circles and draws the crossing without hiding anything, so that no tube is in front, while `twist` and `itwist` cross the two tubes with the second (respectively the first) tube in front; as cobordisms all three are equal. `pair` and `copair` are the bent cylinders (U-turns) that make 2Cob a rigid category. `\PLgenerators` draws them all:

```
\PLgenerators[names]
\PLgenerators*[vertical, names, list={pant, copant, unit, counit, twist, itwist}]
```



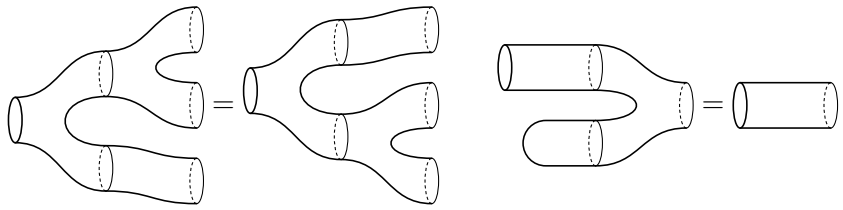
`\PLgenerators[`⟨*options*⟩`]` draws the generators in the key `list` (by default all of them except `twist` and `itwist`) next to each other; the starred form leaves out `id`.

3 The features

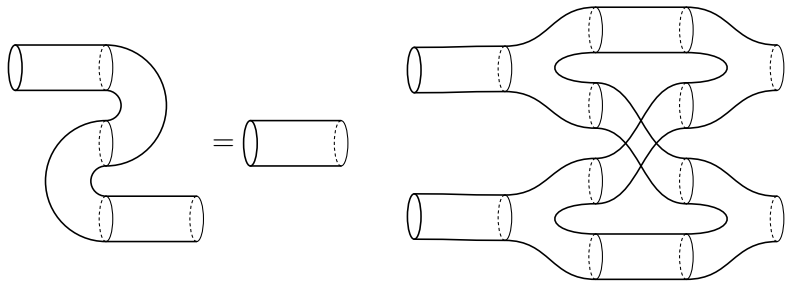
3.1 Composition

Consecutive `\PL` commands form one diagram (spaces and single line breaks between them are allowed; an empty line or any other token ends the diagram). The output circles of one layer are glued to the input circles of the next one, so the counts must agree. Options can be given to any `\PL` of the chain and apply to the whole diagram.

```
\[ \PL{1,2}{pant}\PL{2,3}{pant, id} = \PL{1,2}{pant}\PL{2,3}{id, pant}
   \qquad
   \PL{1,2}{id, unit}\PL{2,1}{copant} = \PL{1,1}{id} \]
```



```
\[ \PL{1,3}{id, copair}\PL{3,1}{pair, id} = \PL{1,1}{id}
   \qquad
   \PL{2,2}{id^2}\PL{2,4}{pant, pant}\PL{4,4}{id, twist, id}\PL{4,2}{copant^2} \]
```

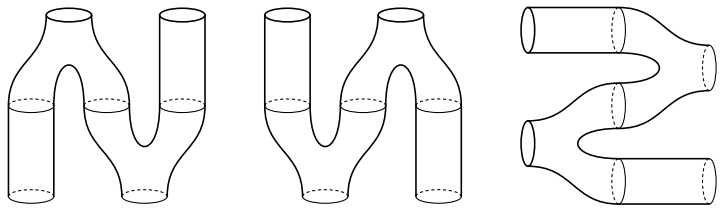


If the circles do not match, or a generator is unknown, the package reports an error that names the offending layer and draws a red ?? instead of the diagram. Should `\PL` already be defined by another package, `\PLlayer` is a synonym with the same arguments (the two can be mixed in a chain).

3.2 Horizontal or vertical

Diagrams are `horizontal` by default: composition runs from left to right and the first circle is on top. With `vertical`, composition runs from top to bottom and the first circle is on the left. `reverse` flips the direction of composition (right to left, or bottom to top). The circles at the start of the diagram are seen from the outside and drawn whole; all other seams show their front half solid and their back half dashed. `orientation=horizontal` and `orientation=vertical` are long forms, and `\plset{`⟨*options*⟩`}` changes the defaults for the rest of the group.

```
\PL[vertical]{2,3}{pant, id}\PL{3,2}{id, copant}
\PL[vertical, reverse]{2,3}{pant, id}\PL{3,2}{id, copant}
\PL[reverse]{2,3}{pant, id}\PL{3,2}{id, copant}
```

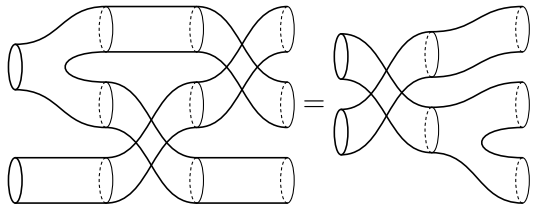


3.3 Reduced form

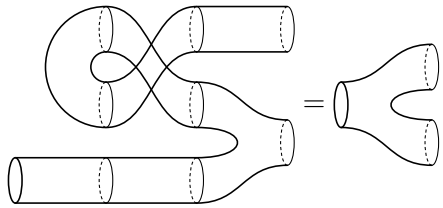
Up to diffeomorphism, a cobordism is determined by which boundary circles lie on the same connected component and by the genus of every component. With `reduce`, the package computes these invariants (by tracking the glued circles and adding up Euler characteristics) and draws the normal form: first the input circles are permuted so that every component gets neighbouring circles, then each component merges its inputs with a tree of `copants`, gets its handles as `pant`/`copant` pairs, and splits into its outputs with a tree of `pants`, and finally the outputs are permuted into place. Components with no inputs start with a `unit`, components without outputs end with a `counit`, and the two kinds of cylinders that bend back (two inputs or two outputs,

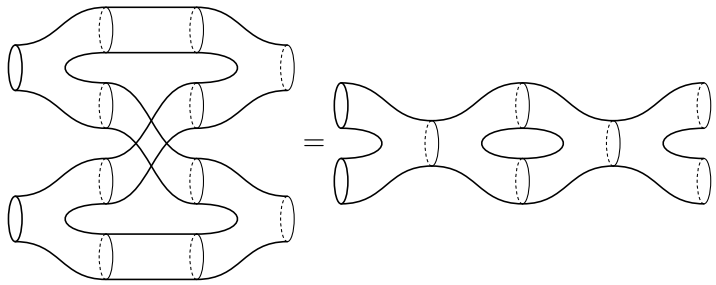
genus 0) become `pair` and `copair`. The permutations are made of swaps; the key `crossing` (`swap`, `twist` or `itwist`) chooses another crossing.

```
\[ \PL{2,3}{pant, id}\PL{3,3}{id, twist}\PL{3,3}{twist, id}
   = \PL[reduce, crossing=twist]{2,3}{pant, id}\PL{3,3}{id, twist}\PL{3,3}{twist, id} \]
```



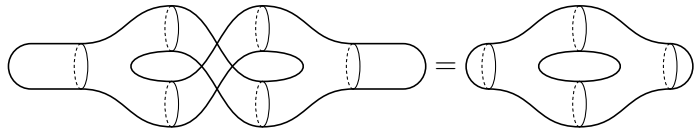
```
\[ \PL{1,3}{copair, id}\PL{3,3}{twist, id}\PL{3,2}{id, copant}
   = \PL[reduce]{1,3}{copair, id}\PL{3,3}{twist, id}\PL{3,2}{id, copant} \]
\[ \PL{2,4}{pant, pant}\PL{4,4}{id, twist, id}\PL{4,2}{copant, copant}
   = \PL[reduce]{2,4}{pant, pant}\PL{4,4}{id, twist, id}\PL{4,2}{copant, copant} \]
```

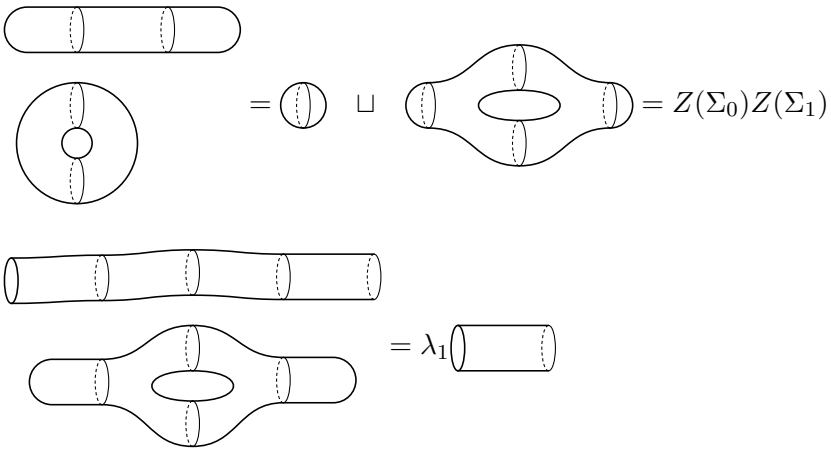




Closed components (surfaces without boundary) are drawn as separate factors, a sphere or a surface with handles, before the rest. With `closed=factor` they are written as numbers instead: a closed surface of genus g evaluates to $Z(\Sigma_g)$ in a TQFT Z, and the key `closed value` sets this expression (`#1` is the genus). The `separator` key puts a symbol between the factors.

```
\[ \PL{0,1}{unit}\PL{1,2}{pant}\PL{2,2}{twist}\PL{2,1}{copant}\PL{1,0}{counit}
   = \PL[reduce]{0,1}{unit}\PL{1,2}{pant}\PL{2,2}{twist}\PL{2,1}{copant}\PL{1,0}{counit} \]
\[ \PL{0,3}{unit, copair}\PL{3,1}{id, pair}\PL{1,0}{counit}
   = \PL[reduce, separator=\sqcup]{0,3}{unit, copair}\PL{3,1}{id, pair}\PL{1,0}{counit}
   = \PL[reduce, closed=factor]{0,3}{unit, copair}\PL{3,1}{id, pair}\PL{1,0}{counit} \]
\[ \PL{1,2}{id, unit}\PL{2,3}{id, pant}\PL{3,2}{id, copant}\PL{2,1}{id, counit}
   = \PL[reduce, closed=factor, closed value={\lambda_{#1}}]
       {1,2}{id, unit}\PL{2,3}{id, pant}\PL{3,2}{id, copant}\PL{2,1}{id, counit} \]
```





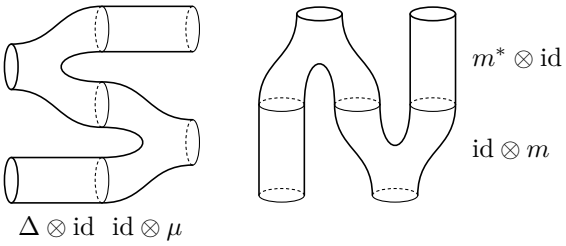
After every diagram (reduced or not) the following macros expand to the invariants of the surface that was just drawn: `\PLlastcomponents` (number of connected components), `\PLlastclosed` (how many of them are closed), `\PLlastgenus` (sum of the genera), `\PLlasteuler` (Euler characteristic) and `\PLlastnormalform` (the normal form as a string of `\PL` commands, closed factors separated by `\cdot`). For the last example: 2 components, 1 closed, genus 1, $\chi = 0$, and the normal form is

```
\PL{0,1}{unit}\PL{1,2}{pant}\PL{2,1}{copant}\PL{1,0}{counit} \cdot \PL{1,1}{id}
```

3.4 Names

The key `names` writes the symbol of each layer under it (or to its right in vertical diagrams). The symbols are set with `symbols`, which takes a key–value list of generator names and math material.

```
\PL[names]{2,3}{pant, id}\PL{3,2}{id, copant}
\PL[vertical, names, symbols={pant=m^*, copant=m}]{2,3}{pant, id}\PL{3,2}{id, copant}
```



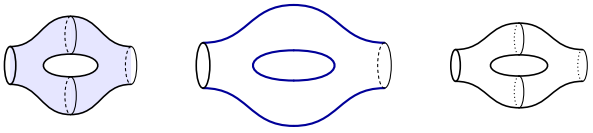
4 Appearance

key	default	meaning
`circle sep`	`1`	distance between neighbouring circles (cm)
`radius`	`0.3`	radius of a circle (cm)
`length`	`1.2`	length of a layer (cm)
`perspective`	`0.3`	width of the ellipses relative to their height
`crotch`	`0.6`	how far the crotch of a pair of pants reaches back
`gap`	`0.4`	space between the factors of a reduced diagram (cm)
`seams`	`true`	draw the circles where layers are glued
`smooth`	`4`	maximal number of layout passes
`scale`	`1`	scales the picture (not the text)
`baseline`	`-0.5ex`	vertical alignment, as in TikZ
`tikz`		further options for the `tikzpicture`
`fill`	`white`	colour of the surface
`opacity`	`0`	opacity of the surface, from `0` (transparent) to `1` (opaque)

By default the surface is fully transparent, so only the outlines are drawn and tubes that pass behind others stay visible. Set `opacity` to a value up to `1` to show the `fill` colour; with `opacity=1` a tube in front hides the outlines behind it.

The drawing uses the TikZ styles `pl surface` (the filled area of every piece), `pl outline`, `pl boundary` (whole circles), `pl seam` (front half of a seam), `pl hidden` (its dashed back half), `pl name`, `pl name right`, `pl separator`, `pl coefficient` and `pl error`. The keys `surface style`, `outline style`, `boundary style`, `seam style` and `hidden style` append to the corresponding styles.

```
\PL[fill=blue!10, opacity=1, length=0.8, circle sep=0.8, radius=0.25]{1,2}{pant}\PL{2,1}{copant}
\PL[seams=false, outline style={thick, blue!60!black}]{1,2}{pant}\PL{2,1}{copant}
\PL[hidden style={densely dotted}, scale=0.7]{1,2}{pant}\PL{2,1}{copant}
```



5 Layout

The numbers of circles change from seam to seam, so their positions are computed. Every generator says where it would like the circles on the neighbouring seams to be: a cylinder keeps the position, a pair of pants puts its legs half a circle distance on either side of the waist, a crossing exchanges its two positions, and generators without inputs (`unit`, `copair`) attach to their neighbours with a weak preference. On every seam the positions are the best least-squares fit to these wishes under the condition that neighbouring circles are at least `circle sep` apart (an isotonic regression), and a few sweeps back and forth through the diagram let both neighbours of a seam have their say. The result is the familiar picture: straight cylinders stay straight and pairs of pants are symmetric whenever possible.

6 The standalone class

With `\documentclass[tikz]{standalone}` every TikZ picture becomes a page of its own. Diagrams inside `\[...\]` and `\(...\)` are handled like in the temperleylieb package: the whole formula becomes

one cropped page. Diagrams outside formulas are pages of their own. Other math environments (`$...$`, `equation`) cannot be turned into pages; the package warns and draws the diagram inline, so use `\[...\]` there, or the class option `varwidth`.

7 Summary of commands

`\PL[`⟨***options***⟩**`]{`**⟨***m,n***⟩**`}{`**⟨***generators***⟩**`}`** One layer; a chain of layers is one diagram. `\PLlayer` is a synonym.

`\PLgenerators*[`⟨***options***⟩**`]`** The generators in `list` side by side.

`\plset{`⟨***options***⟩**`}`** Sets options for the current group.

`\PLlastcomponents`, **`\PLlastclosed`**, **`\PLlastgenus`**, **`\PLlasteuler`** Invariants of the last diagram.

`\PLlastnormalform` Normal form of the last diagram.

The package is written in expl3 and needs a LaTeX kernel from 2020-10 or newer and TikZ. It is distributed under the LaTeX Project Public License 1.3c.